Pure Relative Structured Matrix Learning with Square-Root Matvecs

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Abstract

Let $\mathcal{L} \subseteq \mathbb{R}^{n\times m}$ be any known q-dimensional linear matrix family, and let an unknown A be accessible only through products Ax and $A^\top y$. A recent result obtains a nearly optimal approximation from $\mathcal{L}$ in about $\sqrt{q}$ queries, but incurs factor three and additive error proportional to $\|A\|_F$. Whether sublinear query complexity can give pure $(1+\epsilon)$ relative error was left open.

We resolve this problem with a polynomial-time, nonadaptive algorithm. For a Frobenius-orthonormal basis $B_1, \ldots, B_q$, form the basis-invariant partial trace $S_L = \sum_i B_i B_i^\top$. The algorithm queries the leading eigenspace of S_L exactly from the left, then fits the remaining directions from a Gaussian right sketch. If λ_j are the eigenvalues of S_L, its constant-success query complexity is

$$\min_s \left\{ s + C \begin{bmatrix} \max\{1, \lambda_{s+1}\} \log^2(2q) \\ + \lambda_{s+1}/\epsilon \end{bmatrix} \right\}.$$

The transposed profile is also available. Since $\lambda_{s+1} \leq q/(s+1)$, every family admits pure relative error with $\widetilde{O}(\sqrt{q/\epsilon})$ queries, independent of ambient dimensions and $\|A\|_F/\mathrm{OPT}$. The key proof is a partial-trace covariance lemma for a Gaussian matrix-valued design. Exact high-leverage queries make its expected Hessian the identity, while the centered regression offset has variance only $O(\lambda_{s+1}\mathrm{OPT}^2/k)$. A median construction amplifies success without estimating residual norms. Finally, a symmetric Wishart lower bound for fixed-sparsity approximation yields $\Omega(\sqrt{q/\epsilon})$ adaptive two-sided queries when $q\epsilon$ is bounded below. Thus the universal rate is optimal up to logarithmic factors in this joint regime. We also show that the transformed profile gives pure relative reducible prediction risk under known input and output covariances. For structured positive-definite precisions, it further controls Gaussian KL error and preconditioned condition number.

1 Introduction

Matrix-vector products are the basic access primitive for implicit Hessians, linear predictors, kernel matrices, and matrix functions. When a matrix A is too large to form, one may still apply $x \mapsto Ax$ and, when an adjoint is available, $y \mapsto A^\top y$. This motivates a statistical question: how many such queries reveal the best structured approximation to A?

This question includes a structured prediction problem. If $Y = AX + \eta$ and only input-output experiments with the unknown linear operator are available, then approximating A under the covariance of X is exactly approximating the reducible prediction risk. It also includes matrix-free curvature compression: automatic differentiation supplies Hessian-vector products without materializing a Hessian, while a structured precision can define a scalable Gaussian approximation or preconditioner. Our results apply agnostically in both cases. The target operator need only be near the proposed structure.

We study a known linear matrix family $\mathcal{L} \subseteq \mathbb{R}^{n\times m}$ of dimension q. Given two-sided matvec access to an arbitrary A, the goal is to return $\widehat{B} \in \mathcal{L}$ such that

$$\|A - \widehat{B}\|_F \leq (1+\epsilon) \min_{B\in\mathcal{L}} \|A - B\|_F. \tag{1}$$

The target is agnostic. No stochastic noise model or promise that $A \in \mathcal{L}$ is imposed.

Amsel et al. (2026a) initiated a general theory for this problem. For a finite family $\mathcal{F}$, they give a nearly quadratic improvement from $O(\log|\mathcal{F}|)$ one-sided queries to $\widetilde{O}(\sqrt{\log|\mathcal{F}|})$ two-sided queries. Covering a linear family gives roughly $\sqrt{q}$ queries, but the guarantee is

$$(3+\epsilon)\mathrm{OPT} + \alpha\|A\|_F. \tag{2}$$

They explicitly ask whether every q-dimensional linear family admits pure $(1+\epsilon)$ error with $\widetilde{O}(\sqrt{q}/\operatorname{poly}(\epsilon))$ queries, and whether this can be achieved in polynomial time.

The obstruction is not identifiability. In the realizable case, generic queries often recover a linear family exactly (Halikias and Townsend, 2024). The difficulty is an arbitrary orthogonal residual. A direct Gaussian least-squares fit has an unbiased normal equation, but a few matrix contractions need not embed all q parameter directions. Requiring a conventional subspace embedding returns to order q queries.

Our solution separates the directions that actually obstruct concentration. Let $B_1, \ldots, B_q$ be a Frobenius-orthonormal basis of $\mathcal{L}$ and define

$$S_L = \sum_{i=1}^{q} B_i B_i^\top, \qquad S_R = \sum_{i=1}^{q} B_i^\top B_i. \tag{3}$$

These partial traces are basis invariant, positive semidefinite, and have trace q. We query a leading eigenspace of one partial trace exactly. On its orthogonal complement, Gaussian matvecs have both a well-conditioned Hessian and a small regression offset. The first fact costs the remaining spectral leverage times logarithmic factors. The second costs the same leverage divided by ϵ. Optimizing where to split gives square-root complexity.

Contributions. Our principal results are theoretical.

- We give a nonadaptive, polynomial-time algorithm satisfying (1) for every rectangular linear family. Its query complexity is the better of two basis-invariant spectral profiles, one for each query orientation.
- The profile implies a universal $\widetilde{O}(\sqrt{q/\epsilon})$ upper bound. It has no additive error and no dependence on the ambient dimensions or the ratio $\|A\|_F/\mathrm{OPT}$.
- We prove a Gaussian covariance lemma controlled by a partial trace. The proof combines a dimension-free Schatten moment estimate, an exact fourth-moment calculation, and matrix concentration for unbounded summands.
- We convert the adaptive symmetric Wishart lower bound of Amsel et al. (2026b) into $\Omega(\sqrt{q/\epsilon})$ for general q-dimensional linear families when $q\epsilon = \Omega(1)$. This matches our upper bound up to logarithms. A separate GOE construction shows that $\Omega(1/\sqrt{\epsilon})$ queries can be necessary even for a one-dimensional family.

Table 1: General q-dimensional linear families. Logarithmic factors other than $\log(1/\alpha)$ are suppressed.

Result	Queries	Guarantee
Amsel et al.	$\frac{\sqrt{q\log(1/\alpha)}}{\epsilon^2}$	$(3+\epsilon)\mathrm{OPT} + \alpha\|A\|_F$
This paper	$\sqrt{q/\epsilon}$	$(1+\epsilon)\mathrm{OPT}$

- We lift the theorem to covariance-weighted prediction loss without changing its query count. For positive-definite precision matrices, the same output controls both Gaussian KL divergence and preconditioning quality. Circulant, Toeplitz, and known-graph precision families give explicit dimension-adaptive profiles.

Why two sides matter. The partial trace identifies output directions shared by many basis matrices. A left query reveals one such direction exactly, no matter how many coefficients use it. Gaussian right queries then handle the diffuse remainder. Strong one-sided barriers are known for general matrix-free tasks (Halikias et al., 2026). Our estimator makes the complementary roles of the two orientations explicit.

Table 1 separates the guarantee and query dependence of the closest general result. The earlier linear-family bound is obtained by covering a bounded part of $\mathcal{L}$, then running a finite-candidate procedure. Its runtime scales with the cover size. Our algorithm instead solves linear least squares in the original q coefficients.

2 Why a Spectral Split Is Necessary

It is tempting to fit $\mathcal{L}$ using only a right Gaussian sketch, or only a left sketch, then choose whichever orientation has smaller aggregate variance. This cannot prove a square-root theorem. For an orthonormal basis, define the two offset leverages of a residual $R \perp \mathcal{L}$ by

$$V_R(R) = \sum_i \|R^\top B_i\|_F^2, \qquad V_L(R) = \sum_i \|R B_i^\top\|_F^2. \tag{4}$$

A one-sided Gaussian normal equation has total offset variance proportional to the corresponding quantity. One might hope that $\min\{V_R, V_L\} = O(\sqrt{q}\|R\|_F^2)$, or that their product is at most $q\|R\|_F^4$. Both statements are false.

Proposition 1 (Deleted-tangent obstruction)**.** *For every $a, b \geq 1$, there is a linear family of dimension $q = a + b$ and a unit residual $R \perp \mathcal{L}$ such that*

$$V_R(R) = b, \qquad V_L(R) = a.$$

When a and b are comparable, both global orientations have order-q offset variance. Nevertheless, the spectral-

split estimator removes all contamination with one exact query.

Proof. Let $u, e_1, \ldots, e_a$ and $v, f_1, \ldots, f_b$ be orthonormal systems. Take $R = uv^\top$ and let $\mathcal{L}$ have orthonormal basis

$$\{uf_j^\top : 1 \le j \le b\} \cup \{e_i v^\top : 1 \le i \le a\}. \tag{5}$$

The deleted matrix $uv^\top$ is orthogonal to every basis element. For the first arm, $\|R^\top(uf_j^\top)\|_F = 1$, while the corresponding terms from the second arm vanish. This gives $V_R = b$. The transposed calculation gives $V_L = a$.

The left partial trace is

$$S_L = buu^\top + \sum_{i=1}^{a} e_i e_i^\top.$$

Querying u exactly leaves partial-trace norm one. More strongly, the remaining residual is $(I - uu^\top)R = 0$, so the random-stage offset vanishes. $\square$

This example isolates the role of the deterministic term in (6). It is not included to obtain a conventional subspace embedding. It makes the high-curvature and high-offset directions exact, which leaves a centered problem whose variance is governed by the residual partial trace. A fixed global choice of orientation misses this local structure.

3 Problem and Algorithm

Let $\mathcal{L} \subseteq \mathbb{R}^{n\times m}$ have Frobenius-orthonormal basis $B_1, \ldots, B_q$. Write

$$B_\star = P_{\mathcal{L}} A, \qquad R = A - B_\star, \qquad \mathrm{OPT} = \|R\|_F.$$

Then $\langle R, B\rangle_F = 0$ for every $B \in \mathcal{L}$. For the left partial trace S_L, let $\lambda_1^L \ge \lambda_2^L \ge \cdots \ge 0$ be its eigenvalues, padded by zeros. Fix $s \in \{0, \ldots, n\}$, let P project onto its first s eigenvectors, set $Q = I - P$, and abbreviate $\lambda = \lambda_{s+1}^L = \|QS_LQ\|_{\mathrm{op}}$.

If $\lambda = 0$, query $A^\top$ on an orthonormal basis of $\mathrm{range}(P)$ and minimize $\|P(A-B)\|_F^2$ over $B \in \mathcal{L}$. This recovers $B_\star$ because $QB = 0$ for every $B \in \mathcal{L}$.

Suppose $\lambda > 0$. In addition to the same s left queries, draw $G = [g_1, \ldots, g_k] \in \mathbb{R}^{m\times k}$ with independent standard Gaussian entries and issue the k right queries Ag_ℓ. Return the minimum-norm minimizer

$$\widehat{B} \in \arg\min_{B\in\mathcal{L}} \left\{ \|P(A-B)\|_F^2 + \frac{1}{k}\|Q(A-B)G\|_F^2 \right\}. \tag{6}$$

The objective is observable. The left responses give PA, and QAG is obtained from $AG - PAG$. Equation (6) is an ordinary q-variable least-squares problem.

Define the one-orientation profile

$$\mathsf{Q}_L(\epsilon) = \min_{0\le s\le n} \left\{ s + C\left[\max\{1, \lambda_{s+1}^L\} \log^2(2q) + \frac{\lambda_{s+1}^L}{\epsilon} \right] \right\}, \tag{7}$$

where the bracketed term is zero if $\lambda_{s+1}^L = 0$. Define $\mathsf{Q}_R(\epsilon)$ analogously from S_R. The right profile swaps the roles of A and $A^\top$.

Theorem 2 (Pure relative profile)**.** *For every $A \in \mathbb{R}^{n\times m}$, every q-dimensional linear family $\mathcal{L}$, and every $\epsilon \in (0,1)$, the better orientation of* (6) *returns $\widehat{B} \in \mathcal{L}$ satisfying* (1) *with probability at least* 0.9 *using at most*

$$\min\{\mathsf{Q}_L(\epsilon), \mathsf{Q}_R(\epsilon)\}$$

matvec queries. The query vectors are nonadaptive and the estimator runs in time polynomial in the explicit input size of the basis.

For failure probability δ, multiply only the bracketed random-query term in (7) *by $C\log(1/\delta)$, and likewise for the right profile.*

Computation model. The polynomial-time statement uses real arithmetic and an explicit linearly independent basis. A nonorthonormal basis is whitened through its $q \times q$ Frobenius Gram matrix. The partial trace is then formed explicitly, a symmetric eigenproblem is solved, and (6) is a least-squares problem in q unknowns. No eigengap is required: for any computed rank-s projector, the guarantee holds with the directly certifiable residual leverage $\Lambda_L(P)$ in (9). On the covariance event the least-squares Hessian has smallest eigenvalue at least $1/2$. Bit complexity still depends on the conditioning of the supplied basis, which we do not hide inside the query bound.

The leading eigenspace is not merely convenient. It is the best rank-s choice for this estimator. For an arbitrary rank-s orthogonal projector P, define its residual leverage

$$\Lambda_L(P) = \|(I-P)S_L(I-P)\|_{\mathrm{op}}. \tag{8}$$

The same proof as Theorem 2 gives query count

$$s + C\left[\max\{1, \Lambda_L(P)\} \log^2(2q) + \frac{\Lambda_L(P)}{\epsilon} \right]. \tag{9}$$

Proposition 3 (Optimal exact-query subspace)**.** *Among all rank-s orthogonal projectors P,*

$$\min_{\mathrm{rank}(P)=s} \Lambda_L(P) = \lambda_{s+1}^L.$$

Thus the leading eigenspace minimizes both random-query terms in (9)*.*

Proof. For $Q = I - P$, $\Lambda_L(P)$ is the largest Rayleigh quotient of S_L on range(Q). The claim is the Courant–Fischer characterization of λ_{s+1}^L. □

Corollary 4 (Universal square-root bound)**.** *Let* $L_\epsilon = \log^2(2q) + 1/\epsilon$*. Constant-success pure relative error requires at most*

$$O\Big(\sqrt{qL_\epsilon} + \log^2(2q)\Big) = \widetilde{O}\Big(\sqrt{q/\epsilon}\Big) \tag{10}$$

two-sided matvec queries.

Proof. The partial-trace eigenvalues sum to q, so $\lambda_{s+1}^L \le q/(s+1)$. Also $\max\{1, \lambda\} \le 1+\lambda$. Insert these inequalities into (7) and choose $s = \min\{n, \lfloor\sqrt{qL_\epsilon}\rfloor\}$. If this choice reaches n, the exact queries already cover the row support of $\mathcal{L}$. Otherwise the two s-dependent terms are both $O(\sqrt{qL_\epsilon})$. □

4 Statistical Consequences and Concrete Profiles

The Frobenius objective is not tied to isotropic inputs. Let $\Gamma \succ 0$ and $\Omega \succ 0$ be known input and output weights and define

$$\|M\|_{\Omega,\Gamma}^2 = \|\Omega^{1/2} M \Gamma^{1/2}\|_F^2. \tag{11}$$

Transform the operator and family by $A' = \Omega^{1/2} A \Gamma^{1/2}$ and $\mathcal{L}' = \Omega^{1/2} \mathcal{L} \Gamma^{1/2}$. A right query to A' is implemented by querying A at $\Gamma^{1/2} g$, then multiplying the response by $\Omega^{1/2}$. A left query is implemented analogously. Thus whitening costs no additional oracle calls.

Corollary 5 (Structured prediction risk)**.** *Let* $Y = AX + \eta$*, where* $\mathbb{E}[\eta \mid X] = 0$ *and* $\mathbb{E}[XX^\top] = \Gamma$*. Given two-sided queries to A, the transformed profile of Theorem 2 returns* $\widehat{B} \in \mathcal{L}$ *such that*

$$\begin{aligned}&\mathbb{E}\|\Omega^{1/2}(Y - \widehat{B}X)\|_2^2 - \mathsf{N} \\ &\quad \le (1+\epsilon) \min_{B \in \mathcal{L}} \Big\{ \mathbb{E}\|\Omega^{1/2}(Y - BX)\|_2^2 - \mathsf{N} \Big\},\end{aligned} \tag{12}$$

with probability at least 0.9*, where* $\mathsf{N} = \mathbb{E}\|\Omega^{1/2}\eta\|_2^2$*. The profile uses a basis orthonormal under* (11) *and its corresponding weighted partial traces.*

Proof. The conditional mean assumption gives $\mathbb{E}\|\Omega^{1/2}(Y - BX)\|_2^2 = \mathsf{N} + \|A - B\|_{\Omega,\Gamma}^2$. Apply Theorem 2 to $(A', \mathcal{L}')$ with $\epsilon' = \sqrt{1+\epsilon} - 1 \ge \epsilon/3$, then square its guarantee. Replacing ϵ by ϵ' changes the profile by only a universal constant. □

Table 2: Constant-success profile examples. Tildes hide logarithmic factors.

Family	q	$\|S_L\|_{\mathrm{op}}$	Queries
Circulant	n	1	$\widetilde{O}(1/\epsilon)$
Toeplitz	$2n-1$	H_n	$\widetilde{O}(\log n/\epsilon)$
Graph precision	$n + \|E\|$	$1 + \Delta/2$	$\widetilde{O}((1+\Delta)/\epsilon)$
$a \times b$ block	ab	b	$\min\{a, b\}$ exact

This statement is distribution-free beyond second moments. Gaussian probes are an experimental design chosen by the algorithm, not an assumption on X. The result is therefore an oracle-complexity theorem for the best structured linear predictor under a specified deployment covariance.

For Gaussian uncertainty quantification, the matrix approximation also has a direct distributional consequence.

Corollary 6 (Structured Gaussian precision)**.** *Suppose* $A = A^\top \succeq \mu I$*, every matrix in* $\mathcal{L}$ *is symmetric, and set* $r = (1+\epsilon)\mathrm{OPT}/\mu < 1$*. The output of Theorem 2 is positive definite and satisfies*

$$\mathrm{KL}\Big(N(0, A^{-1}) \,\|\, N(0, \widehat{B}^{-1})\Big) \le \frac{(1+\epsilon)^2 \mathrm{OPT}^2}{4\mu^2(1-r)}, \tag{13}$$

$$\kappa(\widehat{B}^{-1/2} A \widehat{B}^{-1/2}) \le \frac{1+r}{1-r}. \tag{14}$$

Proof. Let $E = A^{-1/2}(\widehat{B} - A)A^{-1/2}$. Then $\|E\|_{\mathrm{op}} \le r$, which proves positive definiteness and (14). If e_i are the eigenvalues of E, the KL divergence is $\frac{1}{2}\sum_i [e_i - \log(1+e_i)]$. For $|e_i| \le r$, the summand is at most $e_i^2/[2(1-r)]$. Finally, $\|E\|_F \le \|A - \widehat{B}\|_F/\mu$. □

Table 2 makes the spectral profile explicit for statistical structures. For real Toeplitz matrices, the normalized diagonal basis gives the ith partial-trace entry $H_n - H_{i-1} + H_{n-1} - H_{n-i}$, so its maximum is H_n. For a known graph precision model, use E_{ii} on vertices and $(E_{ij} + E_{ji})/\sqrt{2}$ on edges. The partial trace is diagonal with entry $1 + \deg(i)/2$. Exact queries can therefore remove hubs before the random stage.

For diagonal matrices, $S_L = I$ and the random stage similarly uses $O(\log^2 q + 1/\epsilon)$ queries, matching specialized dimension-free behavior up to logarithms (Amsel et al., 2026b; Dharangutte and Musco, 2023). For an arbitrary coordinate subspace, the partial traces are row and column degree matrices. Graph degeneracy can give a sharper sparse-only profile (Musco and Ramesh, 2026), while our theorem also covers dense overlapping bases and weighted prediction loss.

Figure 1 is a small falsification check rather than evidence for the theorem. We generated a unit random

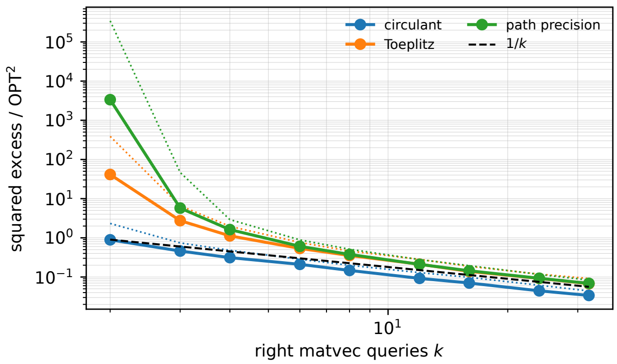

Figure 1: Relative squared excess for three 32×32 families. The dashed line has slope $1/k$.

residual, projected it exactly orthogonal to each family, and solved the right-sketch least-squares problem over 120 independent sketches. The median squared excess follows the predicted $1/k$ behavior once the random design is nonsingular. Dotted curves show the 90th percentile. The supplement gives the complete protocol and seed.

5 Proof of the Upper Bound

We give the main argument and defer moment details and amplification to the supplement. Set $C_i = QB_i$ and define the $q \times q$ Gram matrix

$$K_{ij} = \langle C_i, C_j \rangle_F.$$

Since the original basis is orthonormal, $0 \preceq K \preceq I$. Moreover,

$$\sum_{i=1}^{q} C_i C_i^\top = QS_LQ \preceq \lambda I. \tag{15}$$

For $g \sim N(0, I_m)$, let $D(g) \in \mathbb{R}^{n\times q}$ have columns $C_i g$.

Lemma 7 (Partial-trace Gaussian covariance). *Suppose $K \preceq I$ and $\sum_i C_i C_i^\top \preceq \tau I$ for $\tau \geq 1$. There is a universal constant C such that, if $g_1, \ldots, g_k$ are independent Gaussians and*

$$k \geq C\tau \log^2(2q),$$

then, with probability at least 0.99,

$$\left\| \frac{1}{k} \sum_{\ell=1}^{k} D(g_\ell)^\top D(g_\ell) - K \right\|_{\mathrm{op}} \leq \frac{1}{2}. \tag{16}$$

Proof. The ambient dimensions disappear through Schatten moments. Write $D(g) = \sum_a g_a A_a$, where $A_a = [C_1 e_a, \ldots, C_q e_a]$. Direct calculation gives

$$\sum_a A_a A_a^\top = \sum_i C_i C_i^\top \preceq \tau I, \qquad \sum_a A_a^\top A_a = K \preceq I. \tag{17}$$

Both matrices have trace $\operatorname{tr} K \leq q$. Hence, for every integer $p \geq 1$,

$$\operatorname{tr}\left(\sum_a A_a A_a^\top\right)^p \leq \tau^{p-1} q, \qquad \operatorname{tr} K^p \leq q. \tag{18}$$

Rectangular noncommutative Khintchine (Buchholz, 2001), followed by $\|\cdot\|_{\mathrm{op}} \leq \|\cdot\|_{S_{2p}}$, now gives

$$\left(\mathbb{E}\|D(g)\|_{\mathrm{op}}^{2p}\right)^{1/(2p)} \leq C\sqrt{p\tau}, \qquad p \geq \log(2q). \tag{19}$$

The factors $q^{1/(2p)}$ from (18) are constant. This is the step that removes n and m.

Set $Z = D(g)^\top D(g)$, so $\mathbb{E}Z = K$. For a unit $\theta \in \mathbb{R}^q$, write $C_\theta = \sum_i \theta_i C_i$. Then

$$\theta^\top \mathbb{E} Z^2 \theta = \sum_i \mathbb{E}\langle C_i g, C_\theta g\rangle^2. \tag{20}$$

For a real Gaussian, the identity

$$\mathbb{E}(g^\top M g)^2 = (\operatorname{tr} M)^2 + \operatorname{tr}(M^2) + \operatorname{tr}(MM^\top)$$

and Cauchy–Schwarz yield

$$\begin{aligned} \theta^\top \mathbb{E} Z^2 \theta &\leq \theta^\top K^2 \theta + 2\sum_i \|C_i^\top C_\theta\|_F^2 \\ &= \theta^\top K^2 \theta + 2 \operatorname{tr}\left(C_\theta^\top \sum_i C_i C_i^\top C_\theta\right) \\ &\leq \theta^\top (K^2 + 2\tau K)\theta. \end{aligned} \tag{21}$$

Consequently,

$$\mathbb{E}(Z-K)^2 \preceq 2\tau K \preceq 2\tau I. \tag{22}$$

For independent copies $X_\ell = Z_\ell - K$, the variance parameter of $\sum_\ell X_\ell$ is at most $2k\tau$. Equation (19) and $\|X_\ell\|_{\mathrm{op}} \leq \|D(g_\ell)\|_{\mathrm{op}}^2 + 1$ also imply

$$\left(\mathbb{E}\max_{\ell \leq k} \|X_\ell\|_{\mathrm{op}}^2\right)^{1/2} \leq C\tau \log(2qk). \tag{23}$$

Indeed, take the L_{2p} norm with $p = \lceil \log(2qk) \rceil$ and use the union bound in moment form.

The expected-norm inequality of Tropp (2015) applied to the centered $q \times q$ matrices X_ℓ gives

$$\left(\mathbb{E}\left\|\frac{1}{k}\sum_{\ell=1}^{k} X_\ell\right\|_{\mathrm{op}}^2\right)^{1/2} \leq C\sqrt{\frac{\tau \log(2q)}{k}} + \frac{C\tau \log(2q)\log(2qk)}{k}. \tag{24}$$

For $k = C_0 \tau \log^2(2q)$ and sufficiently large C_0, the right side is at most $1/20$. It decreases for larger k. Markov's inequality proves (16). Supplement A gives the arbitrary-accuracy version. $\square$

The second ingredient controls only the centered offset, which is why the accuracy dependence is $1/\epsilon$ rather than $1/\epsilon^2$.

Lemma 8 (Offset variance)**.** *For any fixed $M \in \mathbb{R}^{n\times m}$, define*

$$\xi_i = \frac{1}{k}\sum_{\ell=1}^{k}\left\{\langle Mg_\ell, C_i g_\ell\rangle - \langle M, C_i\rangle_F\right\}.$$

Under (15),

$$\mathbb{E}\|\xi\|_2^2 \le \frac{2\lambda}{k}\|M\|_F^2. \tag{25}$$

Proof. The real Gaussian quadratic-form identity implies $\mathrm{Var}(\langle Mg, C_i g\rangle) \le 2\|M^\top C_i\|_F^2$. Summing over i gives

$$\sum_i \|M^\top C_i\|_F^2 = \mathrm{tr}\left(M^\top \sum_i C_i C_i^\top M\right) \le \lambda \|M\|_F^2.$$

Independence across the k columns completes the proof. □

Proof of Theorem 2. In orthonormal coordinates of $\mathcal{L}$, the Hessian of (6) is

$$H = H_P + \frac{1}{k}\sum_{\ell=1}^{k} D(g_\ell)^\top D(g_\ell),$$
$$(H_P)_{ij} = \langle PB_i, PB_j\rangle_F.$$

Since $H_P + K = I$, Lemma 7 with $\tau = \max\{1,\lambda\}$ gives $H \succeq I/2$.

At $B_\star$, exact orthogonality $\langle R, B_i\rangle_F = 0$ makes the gradient equal to the centered vector in Lemma 8, with $M = QR$. If d is the coefficient vector of $\widehat{B} - B_\star$, the normal equation is $Hd = \xi$. On the Hessian event,

$$\|\widehat{B} - B_\star\|_F^2 = \|d\|_2^2 \le 4\|\xi\|_2^2.$$

Lemma 8 and Markov's inequality show that $k \ge C\lambda/\epsilon$ makes this at most $2\epsilon\mathrm{OPT}^2$ with constant probability. Finally, orthogonality gives the exact identity

$$\|A - \widehat{B}\|_F^2 = \mathrm{OPT}^2 + \|\widehat{B} - B_\star\|_F^2. \tag{26}$$

Since $\sqrt{1+2\epsilon} \le 1+\epsilon$, the constant-success result follows.

□

We use the metric-median device of Amsel et al. (2026b). Naive confidence amplification would estimate each candidate's residual norm. A Frobenius sketch accurate enough to distinguish $(1+\epsilon)$ error costs $O(1/\epsilon^2)$ extra queries, which would destroy the rate. The following metric argument avoids validation entirely.

Lemma 9 (Validation-free amplification)**.** *Run the Gaussian stage independently an odd number r of times, reusing the same exact responses. For candidate i, let ρ_i be its median Frobenius distance to the r candidates, including itself. Return a candidate minimizing ρ_i. If each candidate satisfies*

$$\|\widehat{B}_i - B_\star\|_F^2 \le \gamma\mathrm{OPT}^2 \tag{27}$$

with probability at least $19/20$, then $r = C\log(1/\delta)$ gives

$$\|\widehat{B} - B_\star\|_F^2 \le 9\gamma\mathrm{OPT}^2$$

with probability at least $1-\delta$.

Proof. A Chernoff bound gives a strict majority of candidates satisfying (27). Condition on this event. Every good candidate has median radius at most $2\sqrt{\gamma}\mathrm{OPT}$. The selected candidate has no larger radius. Its median ball and the strict majority of good candidates must intersect. The triangle inequality through a candidate in the intersection gives distance at most $3\sqrt{\gamma}\mathrm{OPT}$ from $B_\star$. □

Set $\gamma = 2\epsilon/9$ and apply (26). This proves the confidence statement in Theorem 2. Pairwise distances are computed from the known coefficient vectors, so amplification uses no new kind of oracle query.

6 Lower Bounds and Optimality

The square-root dependence is already necessary for exact recovery. Take the family of all $r \times r$ matrices, so $q = r^2$. Under a continuous Gaussian prior, each adaptive left or right query imposes at most r linear constraints. Fewer than r queries leave a non-atomic posterior on a positive-dimensional affine space. Since pure relative error must be exact when $\mathrm{OPT} = 0$, Yao's principle gives $\Omega(\sqrt{q})$ queries.

Accuracy alone also forces a square-root rate, even for one parameter. This endpoint is useful because the fixed-sparsity construction below requires $q\epsilon$ to be bounded below.

Proposition 10 (One-parameter accuracy lower bound)**.** *For every sufficiently small ϵ, there are an ambient dimension d and a one-dimensional family $\mathcal{L}$ such that every possibly randomized and adaptive two-sided algorithm needs $\Omega(1/\sqrt{\epsilon})$ queries for a constant-probability $(1+\epsilon)$ approximation.*

Proof. Let $\mathcal{L} = \mathrm{span}\{I_d/\sqrt{d}\}$ and draw A from the Gaussian orthogonal ensemble. Thus the law is invariant under $A \mapsto U^\top AU$, its diagonal entries have variance two, and its upper off-diagonal entries have variance one. Since $A = A^\top$, transpose queries give no extra access.

Fix a deterministic adaptive algorithm. New queries may be orthogonalized against earlier ones because the response on their span is already known. After t orthonormal queries, rotate them to the first t coordinates. The transcript determines the first t rows and columns, while the bottom-right $(d-t) \times (d-t)$ block remains an independent GOE matrix. This remains true for adaptive queries by sequential conditioning and rotational invariance.

The optimal coefficient is $\beta_\star = \langle A, I_d/\sqrt{d}\rangle_F = \mathrm{tr}(A)/\sqrt{d}$. Conditional on the transcript, its unknown part is Gaussian with variance $2(d-t)/d$. If $t \le d/2$, this variance is at least one. Also $\mathbb{E}\|A\|_F^2 = d(d+1)$. With probability at least 0.9, $\mathrm{OPT}^2 \le \|A\|_F^2 \le 10d(d+1)$. Any successful output must satisfy

$$(\widehat{\beta} - \beta_\star)^2 \le ((1+\epsilon)^2 - 1)\mathrm{OPT}^2 \le 3\epsilon \mathrm{OPT}^2. \quad (28)$$

Take $d = \lfloor c/\sqrt{\epsilon} \rfloor$ for a sufficiently small constant c. On the norm event, the right side of (28) is a small universal constant. Gaussian anti-concentration bounds the conditional success probability away from one for every transcript-measurable estimate. Thus $t > d/2 = \Omega(1/\sqrt{\epsilon})$ is necessary. Yao's principle extends the conclusion to randomized algorithms. Supplement C gives explicit constants and the conditioning induction. $\square$

The dependence on ϵ can be coupled to q. The following result is a reparameterization of the adaptive fixed-sparsity lower bound of Amsel et al. (2026b).

Theorem 11 (Joint lower bound)**.** *There are universal constants $c, c_0, c_1 > 0$ such that, for $\epsilon < c_0$ and $q\epsilon \ge c_1$, there is a q-dimensional linear matrix family for which every possibly randomized and adaptive two-sided algorithm using fewer than*

$$c\sqrt{q/\epsilon}$$

queries fails to return a $(1+\epsilon)$ approximation with constant probability.

Proof. Choose a sparsity level u. The cited theorem constructs a symmetric $d \times d$ Wishart hard instance with $d = \Theta(u/\epsilon)$ for any fixed pattern having between constant multiples of u entries in every row and column. It requires $\Omega(u/\epsilon)$ adaptive forward queries. Symmetry makes a transpose query identical to a forward query.

Choose a cyclic u-regular pattern S. Its coordinate family

$$\mathcal{L}_0 = \{B : B = S \circ B\}$$

has dimension

$$q_0 = du = \Theta(u^2/\epsilon). \quad (29)$$

The source lower bound becomes

$$\Omega(u/\epsilon) = \Omega(\sqrt{q_0/\epsilon}). \quad (30)$$

For $q\epsilon$ above a universal constant, choose $u = \lfloor c_2\sqrt{q\epsilon} \rfloor$ with c_2 small enough. The constants in $d = \Theta(u/\epsilon)$ can be chosen so that $c_3 q \le q_0 \le q$. Integer rounding changes only universal constants.

It remains to reach dimension exactly q. Set $p = q - q_0$, let $\mathcal{D}_p$ be the p-dimensional diagonal family on a disjoint block, and define

$$\mathcal{L} = \{B_0 \oplus D : B_0 \in \mathcal{L}_0,\ D \in \mathcal{D}_p\}.$$

Embed a hard matrix as $A' = A_0 \oplus 0$. A query at (x, z) returns $(A_0 x, 0)$, so it is simulated by one query to A_0. The same holds for a transpose query. If an output $\widehat{B}_0 \oplus \widehat{D}$ succeeds, then

$$\begin{aligned} \|A_0 - \widehat{B}_0\|_F^2 &\le \|A_0 - \widehat{B}_0\|_F^2 + \|\widehat{D}\|_F^2 \\ &\le (1+\epsilon)^2 \|A_0 - S \circ A_0\|_F^2. \end{aligned}$$

Restriction to the first block would therefore solve the source problem. Equation (30) and $q_0 \ge c_3 q$ prove the claim. The supplement records the source theorem and rounding explicitly. $\square$

Theorems 2 and 11 match up to logarithmic factors whenever $q\epsilon = \Omega(1)$. We do not claim a matching lower bound in the edge regime $q\epsilon = o(1)$. The exact-recovery and GOE constructions still give separate square-root lower bounds in q and $1/\epsilon$, while the upper theorem remains valid throughout.

7 Related Work and Discussion

Fixed-sparsity approximation is a coordinate-subspace instance of our problem. Gaussian rowwise regression gives $O(s/\epsilon)$ queries for maximum row degree s, with a matching adaptive lower bound (Amsel et al., 2026b). More recent work characterizes exact sparse recovery by graph degeneracy and gives a nearly matching approximation algorithm (Musco and Ramesh, 2026). These methods exploit disjoint coordinate support. Our partial-trace profile applies to arbitrary dense and overlapping basis matrices.

The split between deterministic high leverage and randomized low leverage has an analogue in row-sampled least squares (Larsen and Kolda, 2022, 2026). In that setting, leverage belongs to scalar rows of one design and a sample reveals one response row. Here one matvec reveals a tensor contraction shared by all q parameters. The partial trace controls the resulting matrix-valued design. Classical active regression needs $\Theta(q/\epsilon)$ scalar labels (Chen and Price, 2019). Sharp sketch-and-solve theory likewise has excess proportional to design rank divided by sketch size (Epperly and Webber, 2026). The two-sided tensor structure is what permits a square-root improvement.

Hierarchical matrix approximation addresses important nonlinear families that are outside our fixed linear-subspace model. The HODLR algorithm of Chen et al. (2025) uses robust recursive peeling and obtains $(1+\epsilon)$ error in $O(k\log^4(n)/\epsilon^3)$ matvecs. The HSS method of Amsel et al. (2026c) obtains an $O(\log(n/k))$ expected squared-error factor with $O(k\log(n/k))$ queries. These specialized guarantees exploit low-rank block geometry that cannot be represented by one q-dimensional linear family. Conversely, they do not give pure relative error for arbitrary dense linear constraints. Recent adaptive matrix-free low-rank work chooses an unknown numerical rank using residual indicators and emphasizes floating-point performance (Smith et al., 2026). It is complementary to our fixed-family agnostic query theorem.

Structured covariance and curvature approximation motivate the weighted and positive-definite consequences in Section 4. We do not claim a sample-based covariance estimator. Corollary 5 instead quantifies how many exact operator experiments recover the best structured prediction rule, while Corollary 6 translates its error into Gaussian and optimization criteria.

Our logarithmic factors come from controlling unbounded Gaussian covariance summands through noncommutative Khintchine and a general expected-norm inequality (Buchholz, 2001; Tropp, 2015). Removing them may require a sharper lower-tail argument tailored to the regularized Hessian. Another open direction is the exact minimax profile outside the regime of Theorem 11. The present result resolves the pure-relative existence and polynomial-time questions while leaving these finer issues.

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